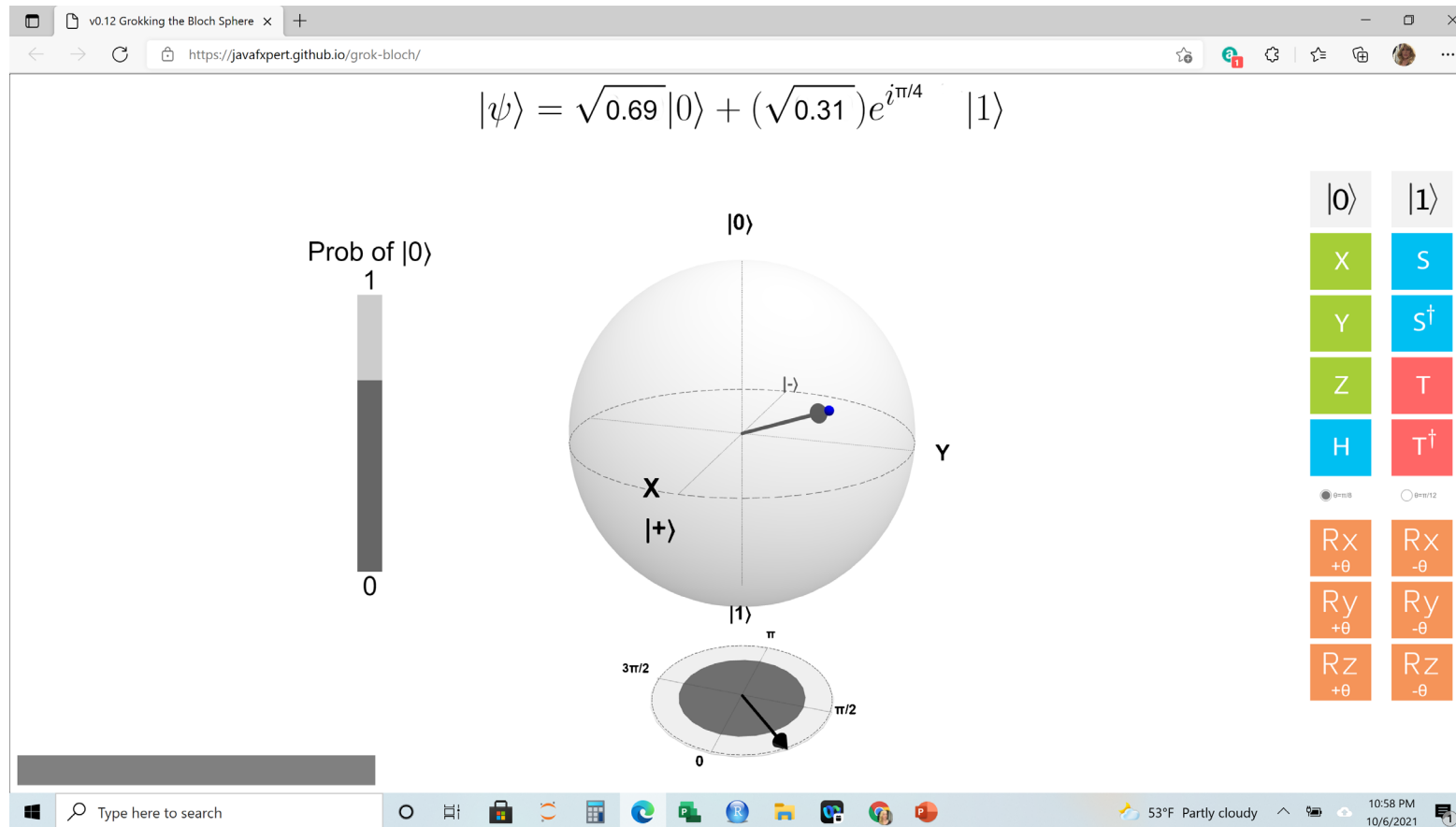


Homework #4 Review (from a previous Class)

Nella Ludlow, PhD
Wright State University

Gronk the Bloch

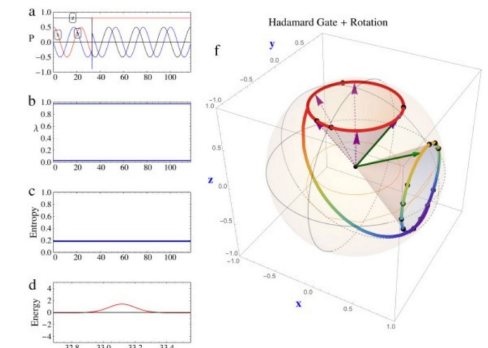
<https://javafxpert.github.io/grok-bloch/>



Quantum Computing - Wright St

The Hadamard gate

While the Pauli gates are a lot like classic logic gates in some respects, the Hadamard gate, or H -gate, is a bona fide quantum beast. It shows up everywhere in quantum computing, and for good reason. The Hadamard gate has the characteristically *quantum* capacity to transform a definite quantum state, such as spin-up, into a murky one, such as a superposition of *both* spin-up and spin-down at the same time.



IBM Quantum Composer

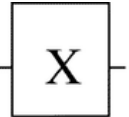
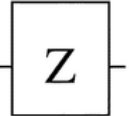
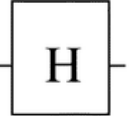
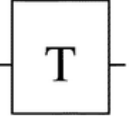
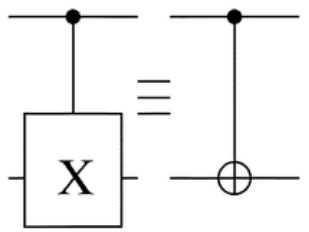
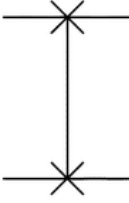
- <https://quantum-computing.ibm.com/composer>

The screenshot displays the IBM Quantum Composer web application. The interface is divided into several sections:

- Composer files:** A sidebar on the left showing a list of files. The first file is "Untitled circuit" with an update time of "a few seconds ago".
- Untitled circuit:** The main workspace for editing a quantum circuit. It includes a toolbar with various quantum gates (H, CNOT, X, Y, Z, T, S, P, RZ, etc.) and a canvas for placing these gates on qubits (q 0, q 1) and classical bits (c 2).
- Visualizations:** At the bottom, there are two visualization tools: "Probabilities" (a bar chart showing the probability of computational basis states) and "Q-sphere" (a Bloch sphere representation of the quantum state).
- Code editor:** On the right, there is a code editor showing OpenQASM 2.0 code. The code includes a header, a qubit declaration, and a measurement operation.

```
1 OPENQASM 2.0;  
2 include "qelib1.inc";  
3  
4 qreg q[2];  
5 creg c[2];  
6  
7 x q[0];
```

Quantum may be the ultimate Matrix Multiplication / Tensor Machine

X Gate Bit-flip, Not		$\equiv$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \beta 0\rangle + \alpha 1\rangle$
Z Gate Phase-flip		$\equiv$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha 0\rangle - \beta 1\rangle$
H Gate Hadamard		$\equiv \frac{1}{\sqrt{2}}$	$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \frac{\alpha+\beta 0\rangle + \alpha-\beta 1\rangle}{\sqrt{2}}$
T Gate		$\equiv$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \alpha 0\rangle + e^{i\pi/4}\beta 1\rangle$
Controlled Not Controlled X CNot		$\equiv$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = a 00\rangle + b 01\rangle + d 10\rangle + c 11\rangle$
Swap		$\equiv$	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = a 00\rangle + c 01\rangle + b 10\rangle + d 11\rangle$

HW4 Review - Q2

ClassMarker - Manage Test

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Question 2

Generic Parent » CS6900 HW4

1 pt

We create this quantum circuit:

Untitled circuit Saved

H ⊕ ⊗ ⊗ X I T S Z T† S† P RZ • |0⟩ ↺ ⓘ ⋮

if ⋮ √X √X† Y RX RY U RXX RZZ + Add

q 0 H

q 1 ⊕

+

c2

Does this produce any qubit into superposition?

✓ A. Yes, qubit 0 is in superposition before measurement.

B. Yes, qubit 1 is in superposition before measurement.

C. Yes, both qubits 0 and qubits 1 are in superposition before measurement.

D. No, none of the qubits are in superposition before measurement.

Question Type: Multiple Choice

Randomize Answers: No

Date Added: Mon 7th Feb 2022

Last Modified: Tue 8th Feb 2022

QID#: 30,359,361

👤 Answers | ✎ Edit | 🔄 Duplicate | 🗣️ Hear In | 🖨️ Printer

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HW4 Review - Q3

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Answers | Edit | Duplicate | Used in | Reorder

Remove from test

Question 3

Generic Parent » CS6900 HW4

1 pt

Untitled circuit Saved

H ⊕ ⊗ ⊗ X I T S Z T† S† P RZ • |0⟩ Z

if ⋮ √X √X† Y RX RY U RXX RZZ + Add

q 0

H

q 1

+

+

c2

If this was run on the Quantum Composer, what state vector outputs will be produced?

A. $|00\rangle$ and $|01\rangle$

B. $|01\rangle$ and $|10\rangle$

C. $|01\rangle$ and $|11\rangle$

✓ D. $|00\rangle$ and $|11\rangle$

Question Type:

Multiple Choice

Randomize Answers:

No

Date Added:

Mon 7th Feb 2022

Last Modified:

Tue 8th Feb 2022

QID#:

30,359,369

HW4 Review - Q4

ClassMarker - Manage Test

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Question 4

Generic Parent » CS6900 HW4

1 pt

Let's use this circuit diagram -- we have nothing on q0, and Hadamard gate H on q1, as shown by:

Untitled circuit Saved

H ⊕ ⊗ ⊗ X I T S Z T† S† P RZ ● |0⟩

if ⋮ √X √X† Y RX RY U RXX RZZ + Add

q 0

q 1 H

+

c2

If we ran this on the Quantum Composer, what are the state vector outputs?

A.

|00> and |01>

B.

|01> and |10>

C.

|10> and |11>

✓ D.

|00> and |10>

Question Type:

Multiple Choice

Randomize Answers:

No

Date Added:

Mon 7th Feb 2022

Last Modified:

Tue 8th Feb 2022

QID#:

30,359,373

HW4 Review - Q5

ClassMarker - Manage Test

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Answers | Edit | Duplicate | Used In | Reorder

Remove From Test

Question 5

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

If we want to rotate the qubit representation around the Z axis (think of the north and south pole), by 180 degrees, or π rotation, what gate would we use?

✓ A.

B.

C.

D.

Question Type:

Multiple Choice

Randomize Answers:

No

Date Added:

Mon 7th Feb 2022

Last Modified:

Mon 7th Feb 2022

QID#:

30,359,376

Answers | Edit | Duplicate | Used In | Reorder

Remove From Test

Question 6

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

HW4 Review - Q6

ClassMarker - Manage Test

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Answers

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Question 6

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

If we want to rotate the qubit representation around the Z axis (think of the north and south pole), by 90 degrees, or $\pi/2$ rotation, what gate would we use?

A. Z gate.

✓

B. S gate.

C. T gate.

D. H gate.

Question Type: Multiple Choice

Randomize Answers: No

Date Added: Mon 7th Feb 2022

Last Modified: Mon 7th Feb 2022

QID#: 30,359,380

Answers

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Question 7

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

HW4 Review - Q7

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Answers

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Question 7

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

If we want to rotate the qubit representation around the Z-axis (think of the north and south pole), by 45 degrees, or $\pi/4$ rotation, what gate would we use?

A.

Z gate.

B.

S gate.

✓ C.

T gate.

D.

H gate.

Question Type:

Multiple Choice

Randomize Answers:

No

Date Added:

Mon 7th Feb 2022

Last Modified:

Mon 7th Feb 2022

QID#:

30,359,383

Answers

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Question 8

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

HW4 Review - Q8

ClassMarker - Manage Test

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Last Modified: Mon 7th Feb 2022

QID#: 30,359,383

👉 Answers

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Question 8

Generic Parent » CS6900 HW4

1 pt

We take a qubit starting at $|0\rangle$ and then we apply a Hadamard gate, which is represented with a point on the equator of the unit sphere.

We want to apply ONE NEW rotation before applying another Hadamard gate to send it to the state of $|1\rangle$.

What gate can provide that rotation?

✓ A.

B.

C.

D.

Z gate.

S gate.

T gate.

H gate.

Question Type: Multiple Choice

Randomize Answers: No

Date Added: Mon 7th Feb 2022

Last Modified: Mon 7th Feb 2022

QID#: 30,359,387

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Question 9

Generic Parent » CS6900 HW4

1 pt

HW4 Review - Q9

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Question 9

Generic Parent » CS6900 HW4

1 pt

We are asked to determine how secure a software lock is. It makes use of the English alphabet of 26 letters. The "key" to open the lock is THREE letters long.
Upper and lower case are considered a different type of input.

What is the MINIMUM number of QUBITS are needed to crack this code using Quantum Parallelism and a code-testing oracle?

(In other words, how many qubits are needed to create a state to represent all the possible combinations using Quantum Parallelism?)

Accepted answers:

- ✓ 18
- ✓ 18.0
- ✓ EIGHTEEN
- ✓ +18
- ✓ +18.0
- ✓ EIGHTTEEN

Question Type: Free Text

Date Added: Mon 7th Feb 2022

Last Modified: Mon 7th Feb 2022

QID#: 30,359,391

👍 Answers | ✎ Edit | 📄 Duplicate | 📌 Used In | ⬆ Reorder

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HW4 Review - Q10

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Question 10

Generic Parent » CS6900 HW4

1 pt

Let's use this circuit diagram -- we have Hadamard gate H on q_0 , and X gate on q_1 as shown by:

Circuit composer

Gates

H S S[†]    X Y Z ID U1 U2 U3 Rx

Ry Rz  T T[†] CH cRz

Operations

|0> if  + Add

Subroutines

Gates overview

Barrier

q[0] |0>

q[1] |0>

q[2] |0>

q[3] |0>

q[4] |0>

+

c5

H

X





































0

1

We assume Identity Matrix H in qubit 0, and Matrix X or NOT in qubit 1, then we can calculate the combined unitary matrix by calculating which of the following?

A.

The 2x2 matrix of H **MATRIX MULTIPLIED** by 2x2 matrix of X **resulting in a 2x2 unitary matrix.**

B.

The 2x2 matrix of H using **TENSOR** with 2x2 matrix of X **resulting in a 2x2 unitary matrix.**

C.

The 2x2 matrix of H **MATRIX MULTIPLIED** by 2x2 matrix of X **resulting in a 4x4 unitary matrix.**

✓

D.

The 2x2 matrix of H using **TENSOR** with 2x2 matrix of X **resulting in a 4x4 unitary matrix.**